

Age and sex breakdowns for ADHD diagnosis in NHS England open Hospital Admitted Patient Care Activity data (2012-2025)

Kaylee Poon

Overview

This analysis explores age, sex, and diagnosis position breakdowns for ICD-10 F90 codes (Hyperkinetic disorders, including ADHD) in publicly available NHS England Hospital Admitted Patient Care Activity data from April 2012 to March 2025.

Methods

Data source

Data extraction methods are based on the [opencodecounts](#) R package. For this analysis we extended the methods to also extract procedural and demographic breakdowns of the counts from publicly available [NHS England Hospital Admitted Patient Care Activity](#) data. Activity is captured in Finished Consultant Episodes (FCEs), where each FCE can have up to 20 recorded ICD-10 diagnoses.

Codelists

A draft ICD-10 codelist¹ including codes that likely indicate an ADHD diagnosis was developed using OpenCodelists.

This codelist is preliminary and may require refinement based on clinical review. The diagnosis codelist includes some codes under review (highlighted in Table 1) where clinical relevance requires further discussion.

¹<https://www.opencodelists.org/codelist/user/kayleepn/adhd-icd-10-test/27569f95/>

Breakdown variables

Diagnostic codes: The following four-character ICD-10 codes were included: Disturbance of activity and attention (F900); *Hyperkinetic conduct disorder* (F901); *Other hyperkinetic disorders* (F908); *Hyperkinetic disorder, unspecified* (F909).

Diagnostic position: Usage counts represent FCEs with a specific code in any diagnostic position (“all diagnoses”) or as the primary reason for admission (“main diagnosis”).

Demographic breakdowns: Demographic breakdowns are available by age groups (with ages 15-19 reported individually and aggregated here for comparability with other age bands) and sex (male, female, unknown).

Results

Diagnostic codes

Table 1 shows that F900 accounts for 97.73% of all F90 code usage. Main diagnoses represent only 0.79% of all F90 diagnosis codes, indicating that ADHD is rarely the primary reason for inpatient admission. Therefore, the following analyses focus on F900 only and do not differentiate between diagnosis positions.

Table 1: ICD-10 F90 codes with total usage by diagnosis position. Main diagnoses as percentage of all diagnoses: F900 <1%; F901 = 8.5%; F909 = 6%. Data source: NHS England [Hospital Admitted Patient Care Activity](#), April 2012 to March 2025.

Code	Description	All diagnoses	Main diagnosis
F900	Disturbance of activity and attention	538,927	3,610
F901	Hyperkinetic conduct disorder	1,686	143
F908	Other hyperkinetic disorders	649	0
F909	Hyperkinetic disorder, unspecified	10,183	612

Age breakdowns

Figure 1 shows F900 usage by age group. All age groups show upward trends, with steep increases from approximately 2021 onwards. Age groups 5-9 through 25-29 show a notable decline during 2020-2021.

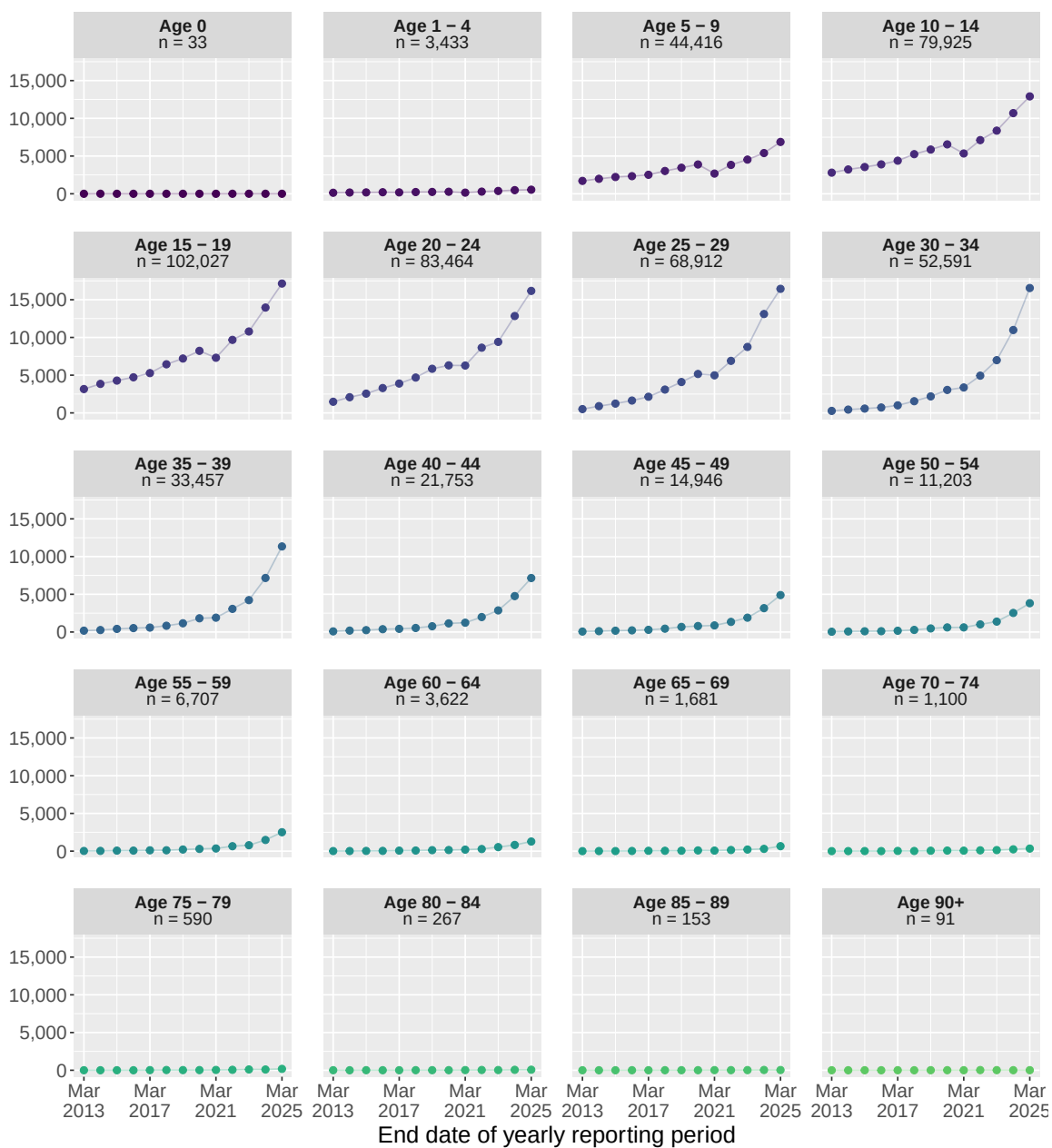


Figure 1: ICD-10 F900 usage over time by age group (ages 15-19 aggregated). Data source: NHS England [Hospital Admitted Patient Care Activity](#), April 2012 to March 2025.

Sex breakdowns

Figure 2 shows F900 usage by sex across all diagnosis positions. Male diagnoses exceeded female diagnoses until approximately 2024, when female diagnoses surpassed male. Gender unknown remains low throughout the period.

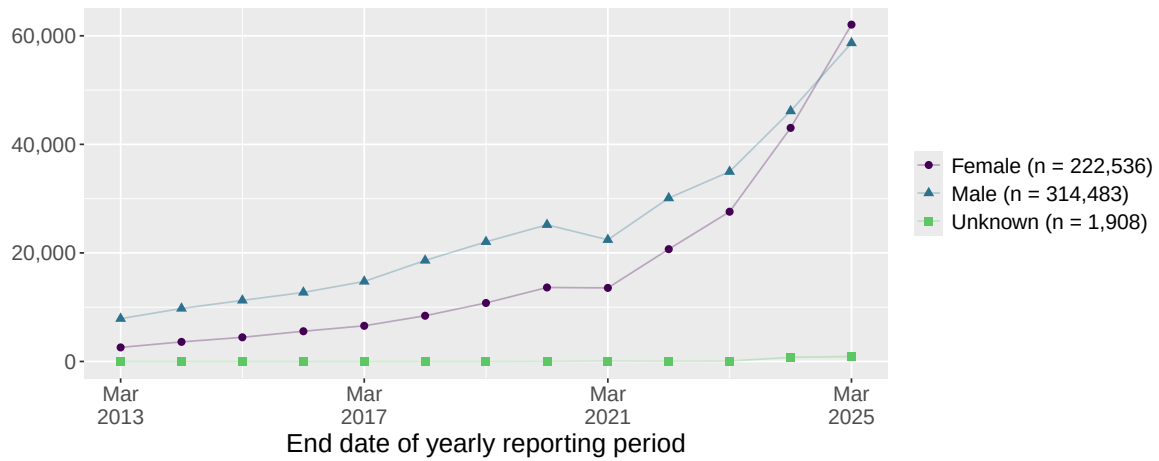


Figure 2: ICD-10 F900 usage over time by sex (all diagnosis positions). Data source: NHS England [Hospital Admitted Patient Care Activity](#), April 2012 to March 2025.